

KAUSTUBH DWIVEDI

Aerospace Structures & Design Engineer

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PROFILE

Aerospace structures and design engineer completing an MSc at Cranfield, with two projects carried out with GKN Aerospace: an LPBF redesign of an Airbus A320 slat-track can, taken from load definition to released native CAD at 20.6 % below the incumbent mass, and an SPH-FEA bird-strike assessment of composite leading-edge concepts validated against manufactured specimens. First-author publication in Materials Today Communications. I work to a stated load basis and a documented evidence chain, and I am explicit about what a result clears and what it does not.

EXPERIENCE

Airbus A320 Slat-Track Can — LPBF Redesign and Structural Substantiation

2025 – 2026

Supervised and reviewed by GKN Aerospace · MSc research project, Cranfield University

- **Released a monolithic AlSi10Mg LPBF design** at 1.478 kg in place of a multi-part welded 6061-T6 assembly, removing 20.6 % of the mass with all five load cases elastic and a minimum yield factor of safety of 5.56.
- **Quantified what the redesign actually bought:** at a fixed 1.450 kg, redistributing wall material alone cut displacement 3.4–5.1 % and von Mises stress 3.7–10.1 %, and a uniform 1.50 mm wall of comparable mass yielded locally where the variable wall stayed elastic.
- **Set the load basis with the industrial supervisor** where the certified definition was proprietary — a five-case directional pressure matrix on CS 25.963 fuel-head loading, bounded against CS 25.561 emergency-landing inertia.
- **Down-selected the manufacturing route** across four AM candidates against the welded incumbent, rejecting both polymer routes on quantified stiffness grounds rather than on preference, and carrying the variable-wall method forward to the metal design.
- **Released the geometry as a watertight native CATIA V5 solid**, exchanged through STEP and re-verified on an independent 239,329-element C3D10 mesh sharing no nodes with the development model.
- **Screened the incumbent for fatigue** to EN 1999-1-3 detail category 50–4.3 in nCode: 1.145×10^6 repeats and a 3.82× margin on the 300,000-repeat target, cross-checked to within 3 % by an independent shell tangent-stress route.
- **Assessed manufacturing risk** with ANSYS Additive thermal simulation, identifying the root and collar band as the sustained hotspot and establishing from a control run on the incumbent geometry that the pattern follows build direction, not the redesign.
- **Closed the work out against a requirements verification matrix**, separating what the numerical evidence closes from what still needs supplier allowables, a buckling check and leak/proof testing before prototype release.

Composite Wing Leading Edge — Bird-Strike Concept Assessment

2025 – 2026

GKN Aerospace / Cranfield University · industry-linked group design project · technical lead

- **Ranked five leading-edge concepts** — CFRP, GFRP, Kevlar/epoxy, honeycomb sandwich and metallic — under common loading, using Abaqus/Explicit SPH-FEA at 90 and 180 m/s soft-body impact on deformation, peak force, stress distribution and energy absorption.
- **Validated the models against manufactured specimens** and established where the modelling approach stopped being reliable, rather than reporting agreement uncritically.
- **Converted the analysis into design recommendations**, weighing laminate behaviour and manufacturability alongside impact performance.
- **Ran the technical coordination:** task allocation, deadline tracking, integration of teammates' analysis and test inputs, and presentation of recommendations at project reviews.

Engineering Design Intern — La Fondation Dassault Systèmes / Dassault Systèmes

Dec 2022 – May 2024

Pune, India · 18-month industry placement

- **Took a modular multi-ratio gearbox from concept to working prototype**, combining CATIA V5 / 3DEXPERIENCE modelling, hand calculation and Abaqus assessment.
- **Iterated the design against load transfer, packaging and manufacturability constraints**, keeping the technical documentation current with each change.
- **Worked to a structured review cadence** inside a Dassault Systèmes engineering environment, responding to senior-engineer feedback and managing design changes through to sign-off.

TECHNICAL SKILLS

Finite element	Abaqus/Standard and /Explicit — shell and solid modelling, geometric and material nonlinearity, contact and tie constraints, SPH impact, mesh sensitivity studies, verification against an independent mesh, simulation–test correlation.
Optimisation & AM	Tosca Structure (bead and shape), ANSYS Additive thermal simulation; DfAM — wall thickness, build orientation, support strategy, process risk.
CAD & release	CATIA V5 and 3DEXPERIENCE — part, assembly and multi-section surface modelling, native solid reconstruction, STEP exchange, geometry validation and design documentation.
Composites	Laminate and sandwich design, stacking sequence definition, specimen manufacture, mechanical and impact testing, damage and delamination modelling.
Standards & method	CS 25.963 and CS 25.561 load definitions; EN 1999-1-3 fatigue curves; nCode; requirements verification matrices and evidence chains.
Programming	Python for post-processing, mesh and field-data handling and figure generation; MATLAB; Excel.

EDUCATION

MSc Advanced Lightweight and Composite Structures — Cranfield University Accredited by RAeS and IMechE for further learning toward CEng registration. Composite structures, finite element methods, structural stability, impact mechanics. <i>Thesis: Slat Track Cans of the Future — Exploring the Design Space of a High Criticality Fuel-Tank Component.</i>	Sept 2025 – Sept 2026
B.Tech Mechanical Engineering — Symbiosis Institute of Technology, Pune CGPA 8.36 / 10.	2020 – 2024

PUBLICATIONS AND RESEARCH

Dwivedi, K., et al. (2024). Optimizing 3D Printed Diamond Lattice Structure and Investigating the Influence of Process Parameters on their Mechanical Integrity Using Nature-Inspired Machine Learning Algorithms. *Materials Today Communications*, 38, 108233. doi:10.1016/j.mtcomm.2024.108233

Co-author of three peer-reviewed CRC Press book chapters on polymer welding and process optimisation. doi:10.1201/9781003437369

- Designed, manufactured and compression-tested 3D-printed PLA+ diamond lattice specimens, linking process parameters and cell geometry to mechanical performance.

MEMBERSHIP AND STATUS

- Royal Aeronautical Society — Student Affiliate. Issued Nov 2025, membership ID 3072018.
- UK Graduate Route eligible on MSc completion. Open to UK relocation, hybrid working and travel to client or manufacturing sites.